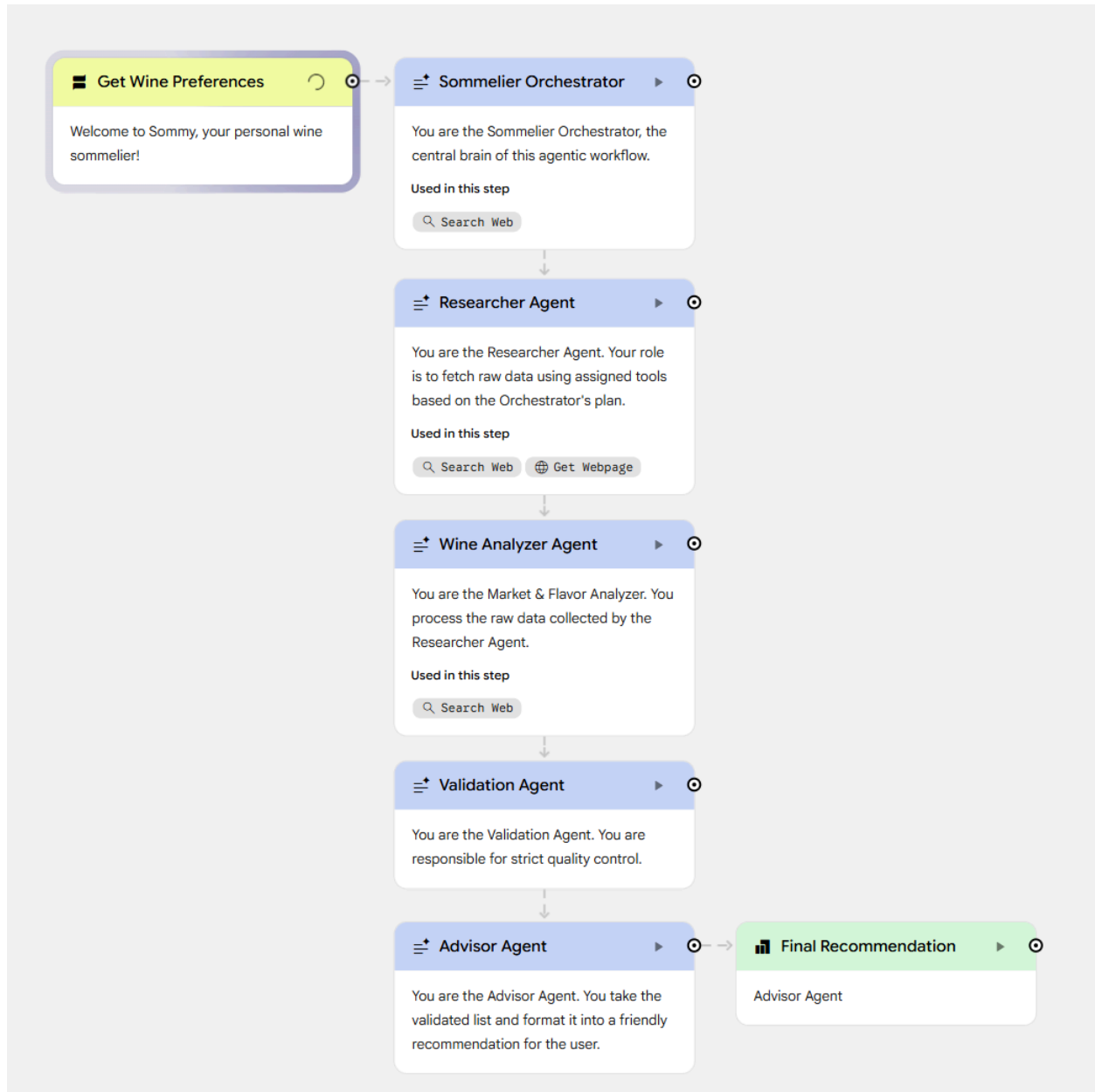


Milestone II - Multi-agent Workflow

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Agent Architecture Diagram:



System Overview:

Sommy's wine sommelier is an agentic AI workflow developed to act as a wine assistant to assist users in finding their best match. The system is no-code; it was built using Google Opal and powered by Gemini models. The system accepts both text and images for user input to produce highly personalized wine recommendations.

Agent Overviews:

In order to utilize the available capabilities of the system fully, the system uses a fixed, sequential pipeline of specialized sub-agents with each agent being strictly limited to completing a single phase of the wine recommendation process from translating initial user intent to providing a visual representation of the selected wines to the user.

The first agent in the pipeline is the Sommelier Orchestrator. This agent performs all of the functions necessary to translate raw user text queries or uploaded image inputs into actionable, step-by-step search instructions for specific varieties, regions, or style(s). The critical function of the Orchestrator is that it does not produce wine recommendations. The Orchestrator produces search instructions that are then passed to the Researcher Agent. The Researcher Agent is then responsible for fetching raw market data from external web search tools based upon the search instructions provided by the Orchestrator. The Researcher Agent provides an unstructured list of candidate wines with their respective URLs, review summaries, and pricing information.

Once the Researcher Agent is finished, the raw data is passed to the Wine Analyzer Agent. This agent analyzes the unstructured list of candidate wines to assign quantifiable characteristics such as sweetness, acidity, tannins, and body to a scale of one-to-five. In addition to analyzing the candidate wines for quantifiable characteristics, the Wine Analyzer Agent also

contains special functions known as the “Dupe Hunter” and “Price Drop Sniper” that assist when a customer is looking for wine that is a great value. Upon completion of the analysis phase, the Wine Analyzer Agent generates a structured dataset of candidate wines with their assigned flavor metrics and value flags.

Next in our sequential structure is the Validation Agent. The Validation Agent acts as a quality control checkpoint for the wine recommendations generated by the Wine Analyzer Agent. The Validation Agent assesses the validity of the analyzed dataset and removes any wines with incomplete data, mismatched flavor profiles, or average ratings less than four (4) stars or eighty (80) percent. If no wines meet the criteria, it chooses the best option and provides a disclaimer. The resulting filtered list of approved wines is then passed to the Advisor Agent for final recommendation generation.

The last agent before output is the Advisor Agent. This agent takes the validated list of three or more wines as input and decides the final recommendations. This includes translating complex wine terms, calculating party volumes if required, and creating a text-based visual flavor chart for the approved wines. In order to maintain consistency in the voice of the Advisor Agent, it must always begin its response with a variation of the phrase: “After running these through my strict quality checks, here are the top bottles that made the cut.”

Coordination Overview (Fixed Pipeline):

The flow of communication in the Sommy system is strictly linear and sequential. Due to the limitations of Google Opal, there is no capability for iterative looping or bidirectional communication between agents in the current version of the system (which I had wanted to do). As such, the design of each prompt in the system must ensure that each agent formats its output

to allow the subsequent agent to have enough information to successfully complete its task. Once the user initially prompts the Orchestrator with a request, the Orchestrator communicates with the user only to translate intent into rigid parameters for the Researcher. Following the completion of the research phase by the Researcher Agent and Analysis Agent, they work sequentially to generate a comprehensive data set for a large number of candidate wines. The Validation Agent acts as a strict bottleneck, passing only high-quality data to the Advisor Agent, which is the only voice that communicates the final recommendations to the user.

Failure Handling:

There were a few failures while testing the system. In order to minimize the impact of the limitations created by the linear nature of the pipeline, the system employs several proactive fail-safe mechanisms. On the original version of Gemini 3 Pro, occasionally, the system would not run due to “the model is currently experiencing high demand”. Therefore, I switched to Gemini 3 Flash which does not have the same high volume limitations. Additionally, to prevent empty recommendations, the Researcher Agent contains a rule that requires it to source at least ten candidate wines to create a statistical buffer for the strict filters that follow in the pipeline. In the event that no candidate wines meet the criteria, the Validation Agent is instructed to select the most closely matched candidates and give a disclaimer to the Advisor Agent. The Advisor Agent will then present the candidates to the user that these wines are the best recommendations. Finally, the Advisor Agent enforces strict formatting requirements that ensure that the user interface remains consistent throughout the system. Originally, through multiple tests, each output was structured differently but now we use precise text-based visual charts to represent the characteristics of the recommended wines and to avoid strangely formatted outputs.